

REQUEST FLOW



compose.yml | SEE THE PATTERN, NOT REPETITION

services:

frontend:

```
image: nginx:alpine          # image OR build
ports: ["8080:80"]           # HOST:CONTAINER
volumes: ["/usr/share/nginx/html:ro"]
depends_on: [backend]
```

backend:

```
build: ./backend              # folder with Dockerfile
expose: ["5000"]              # internal only
environment:
  DB_HOST: db                  # service name = DNS
  DB_NAME: appdb
  DB_USER: appuser
  DB_PASSWORD: ${DB_PASSWORD} # .env / secret
depends_on: {db: {condition: service_healthy}}
healthcheck:
  test: ["CMD", "curl", "-f", "http://localhost:5000/health"]
restart: unless-stopped
```

db:

```
image: postgres:16-alpine
environment:
  POSTGRES_DB: appdb
  POSTGRES_USER: appuser
  POSTGRES_PASSWORD: ${DB_PASSWORD}
volumes: ["db_data:/var/lib/postgresql/data"]
healthcheck:
  test: ["CMD-SHELL", "pg_isready -U appuser -d appdb"]
```

```
networks: {default: {name: app_net}} # shared service DNS
volumes: {db_data: {}}               # persistent data
```

ESSENTIAL COMMANDS

docker compose up -d

start stack

docker compose up -d --build

rebuild + start

docker compose ps

status / ports

docker compose logs -f

follow all logs

docker compose logs -f backend

one service logs

docker compose exec backend sh

shell into service

docker compose restart backend

restart one service

docker compose config

validate final YAML

docker compose pull

pull newer images

docker compose down

stop + remove stack

docker compose down -v

DANGER: delete volumes

REMEMBER THIS PATTERN

image / build existing image OR your Dockerfile

ports host access: HOST:CONTAINER

expose internal container port

environment runtime config; secrets out of Git

depends_on order; healthcheck adds readiness

volumes persistent data / host files

networks service names become DNS names

WHEN IT BREAKS

ps -> logs -> exec -> curl -> config

localhost trap: backend reaches DB at "db", not localhost

data trap: down -v deletes named-volume data